

THE FOURTH QUADRANT, THE SOLUTION TO THAT MOST USEFUL OF PROBLEMS*

Did Aristotle walk slowly?—Will they follow the principles?—How to manufacture a Ponzi scheme and get credit for it

It is much more sound to take risks you can measure than to measure the risks you are taking.

There is a specific spot on the map, the Fourth Quadrant, in which the problem of induction, the pitfalls of empiricism come alive—the place where, I repeat, absence of evidence does not line up with evidence of absence. This section will allow us to base our decision on sounder epistemological grounds.

David Freedman, RIP

First, I need to pay homage to someone to whom knowledge has a large debt. The late Berkeley statistician David Freedman, who perhaps better than anyone uncovered the defects of statistical knowledge, and the inapplicability of some of the methods, sent me a farewell gift. He was supposed to be present at the meeting of the American Statistical Association that I mentioned earlier, but canceled because of illness. But he prepared me for the meeting, with a message that changed the course of the Black Swan idea: be prepared; they will provide you with a certain set of self-serving arguments and you need to respond to them. The arguments were listed in his book in a section called “The Modelers’ Response.” I list most of them below.

The Modelers’ Response: We know all that. Nothing is perfect. The assumptions are reasonable. The assumptions don’t matter. The assumptions are conservative. You can’t prove the assumptions are wrong. We’re only doing what everybody else does. The decision-maker has to be better off with us than without us. The models aren’t totally useless. You have to do the best you can with the data. You have to make assumptions in order to make progress. You have to give the models the benefit of the doubt. Where’s the harm?

This gave me the idea of using the approach “This is where your tools